



Comprar

A



Comprar

B



Probabilidades de Estados

Comprar

C



Comprar

A



40%

30%

Comprar

B

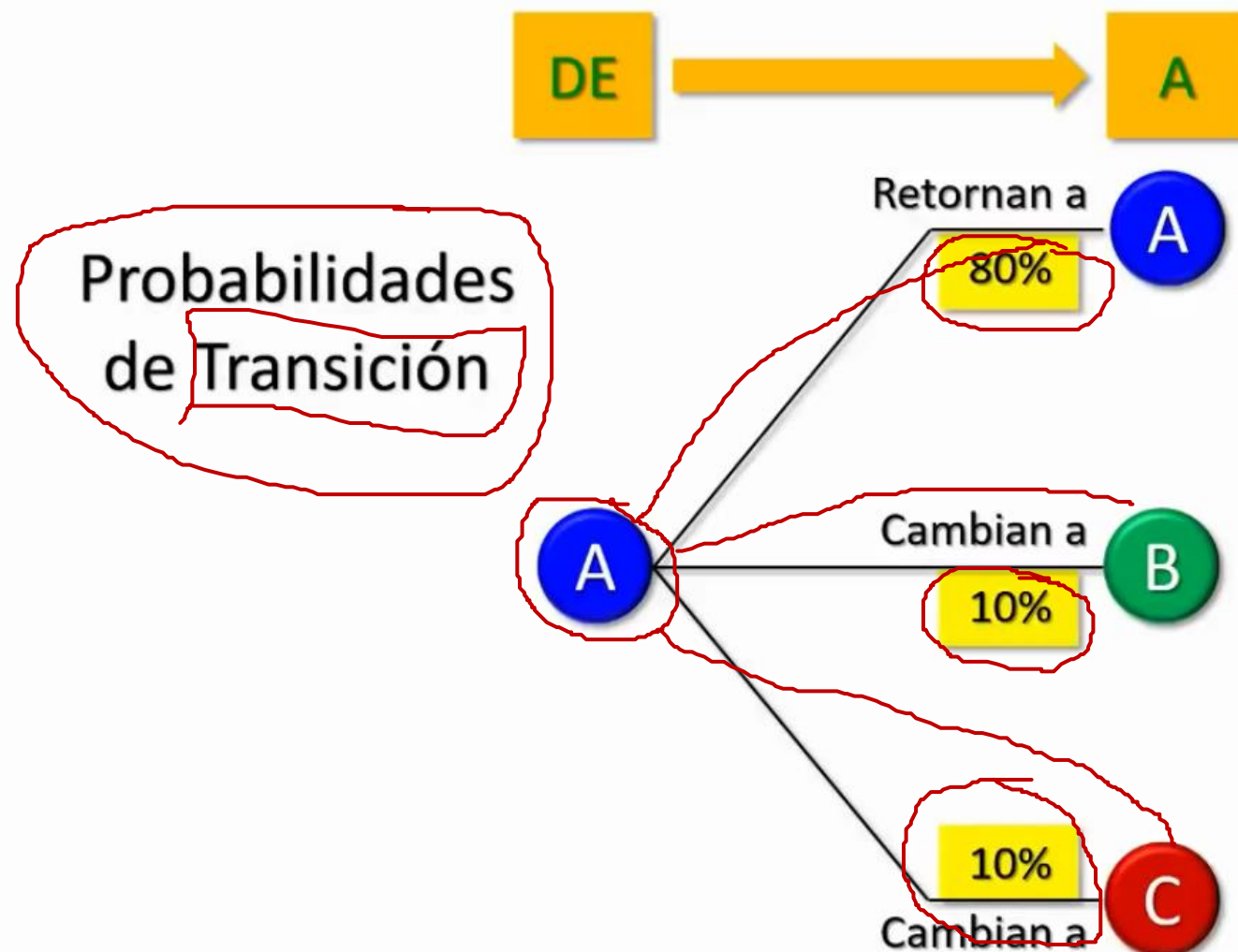


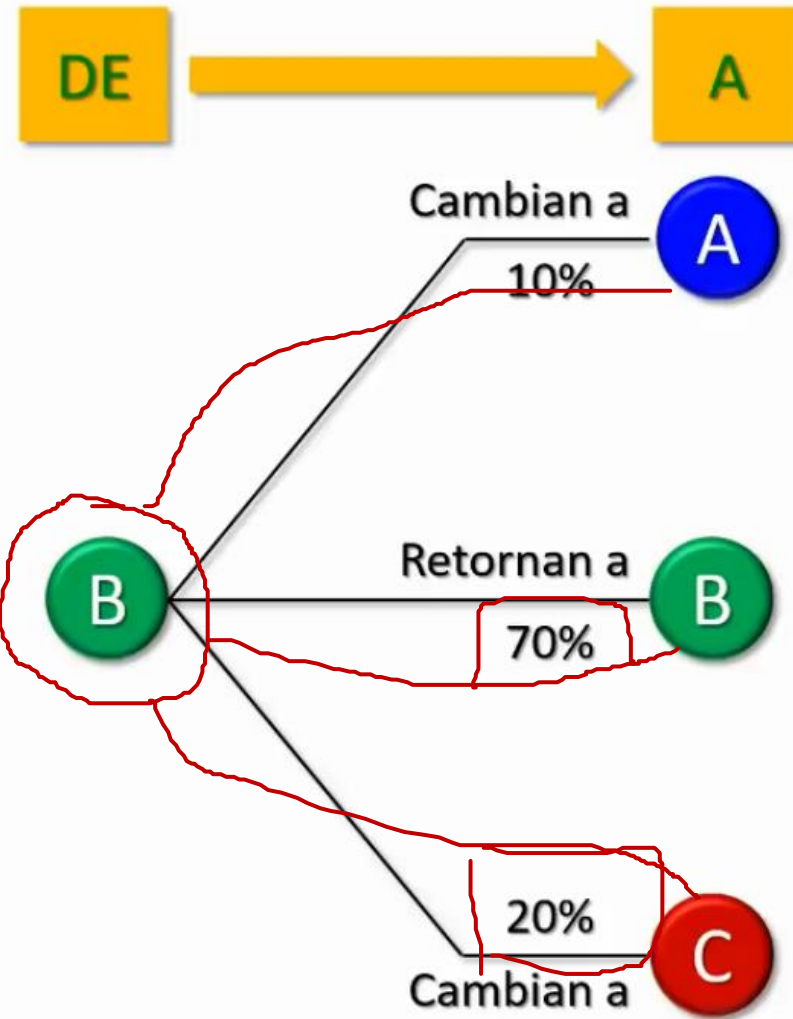
Comprar

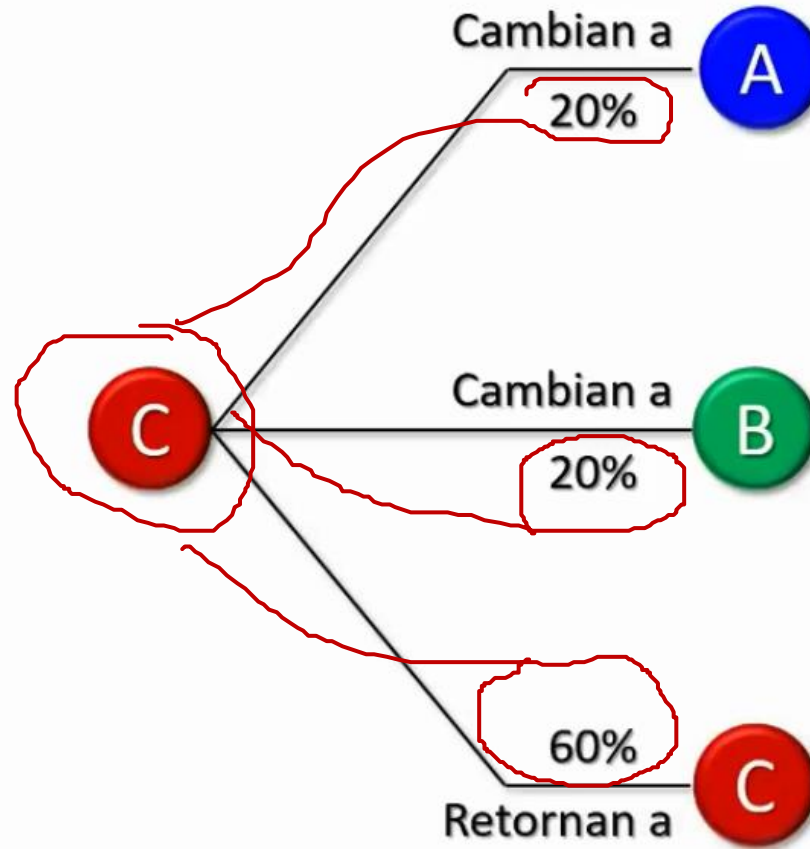
C

30%

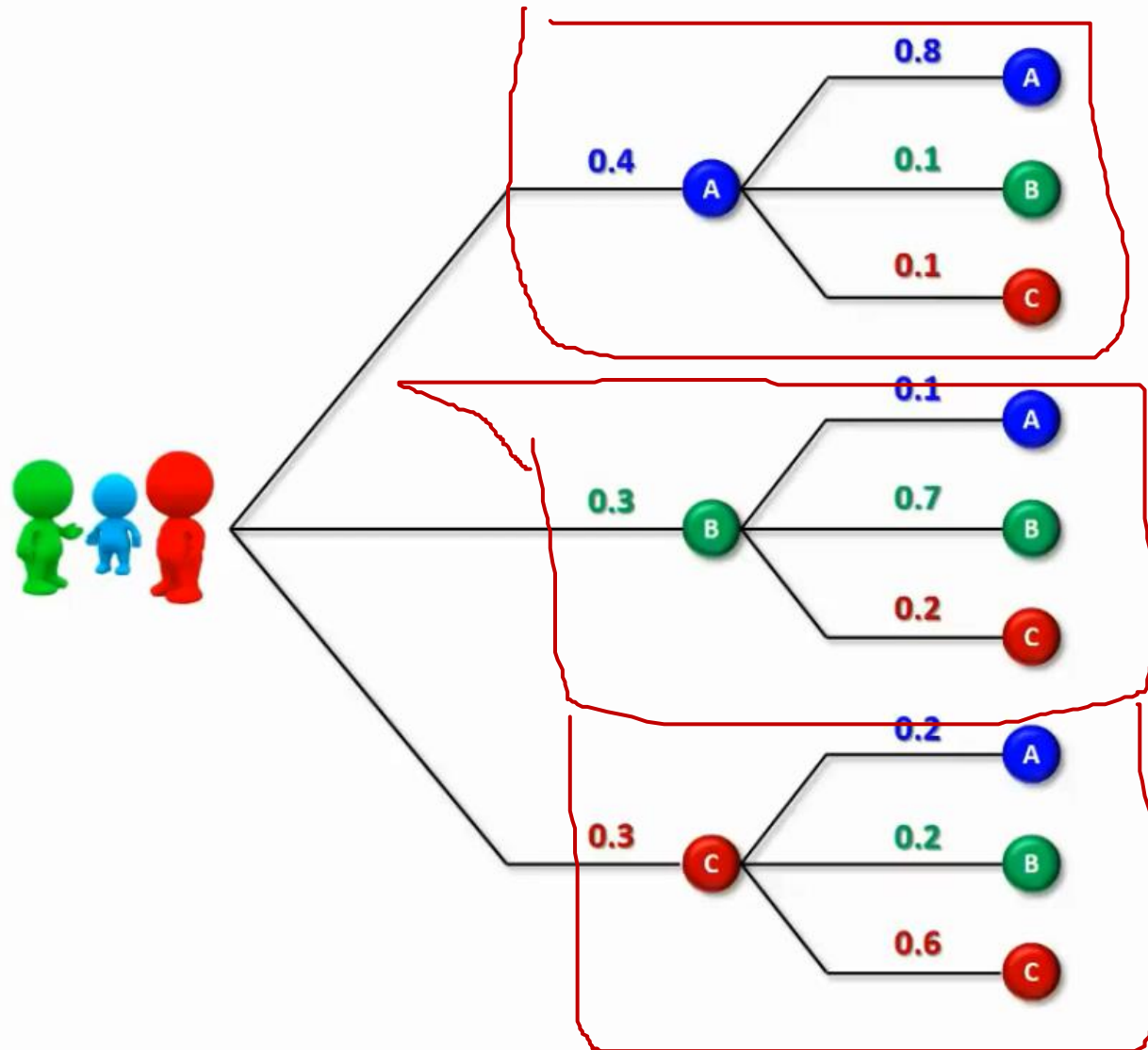




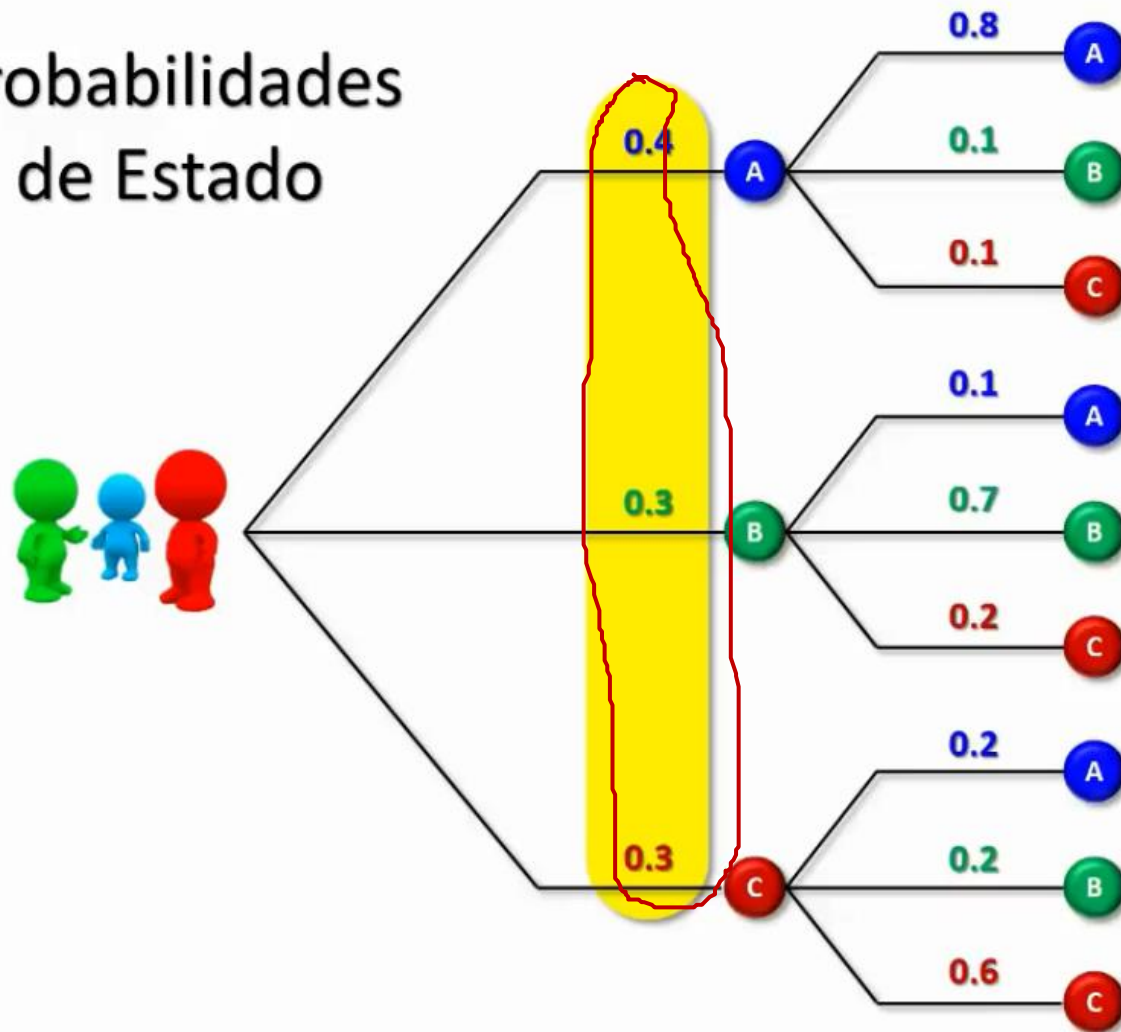




1000 | 0



Probabilidades de Estado

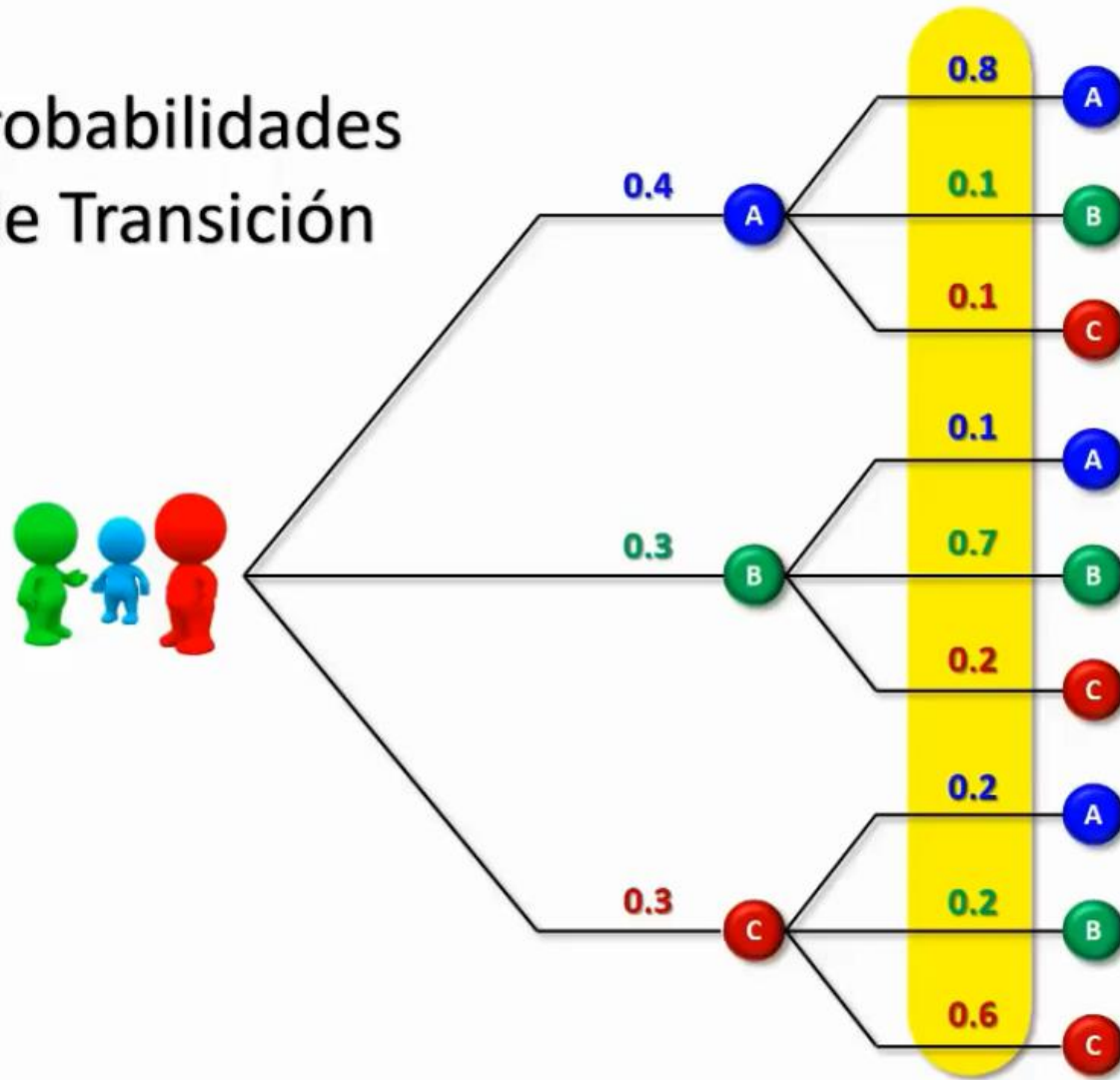


Estado 0

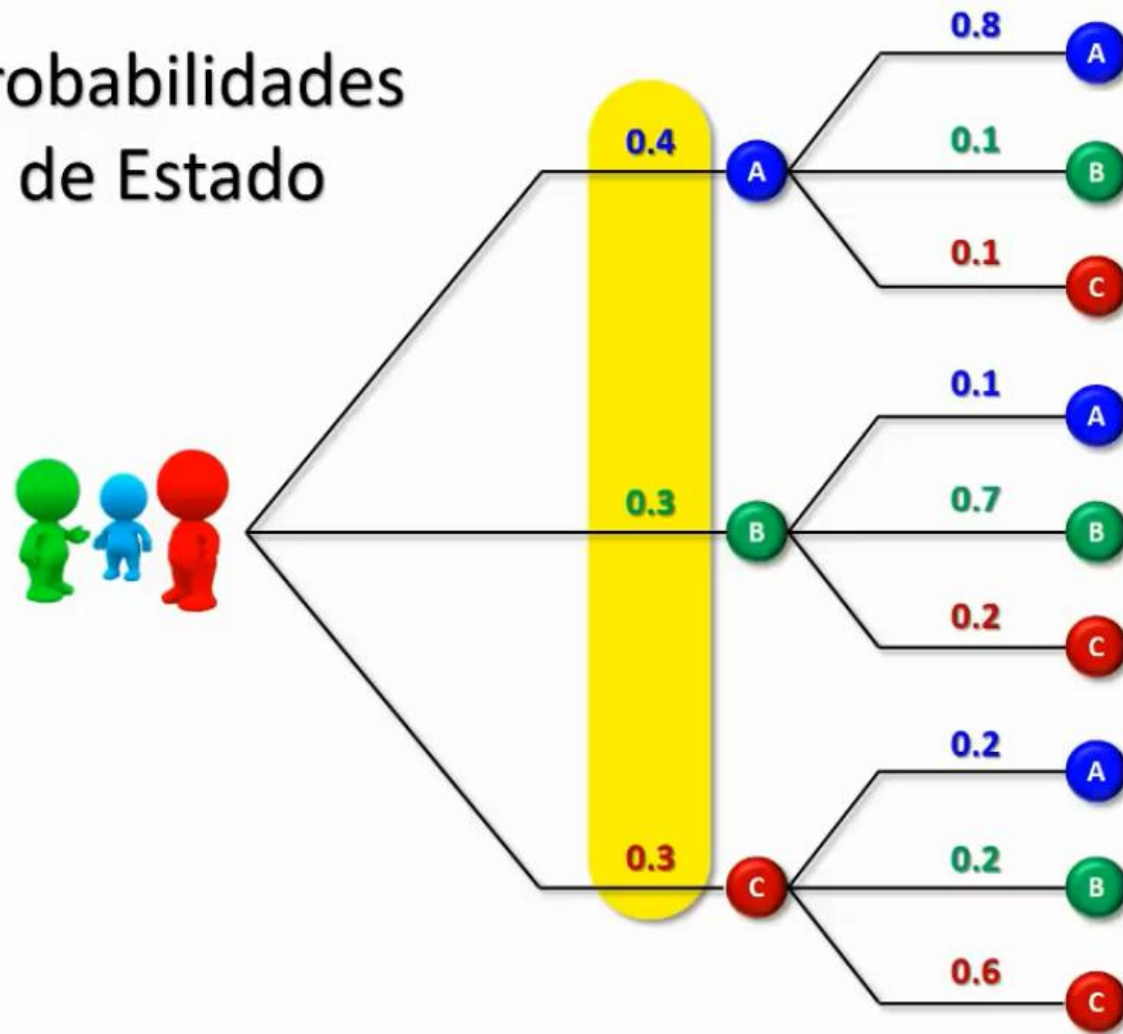
$$\pi(0) = [0.4 \quad 0.3 \quad 0.3]$$

$$\sum \pi_i(0) = 1$$

Probabilidades de Transición



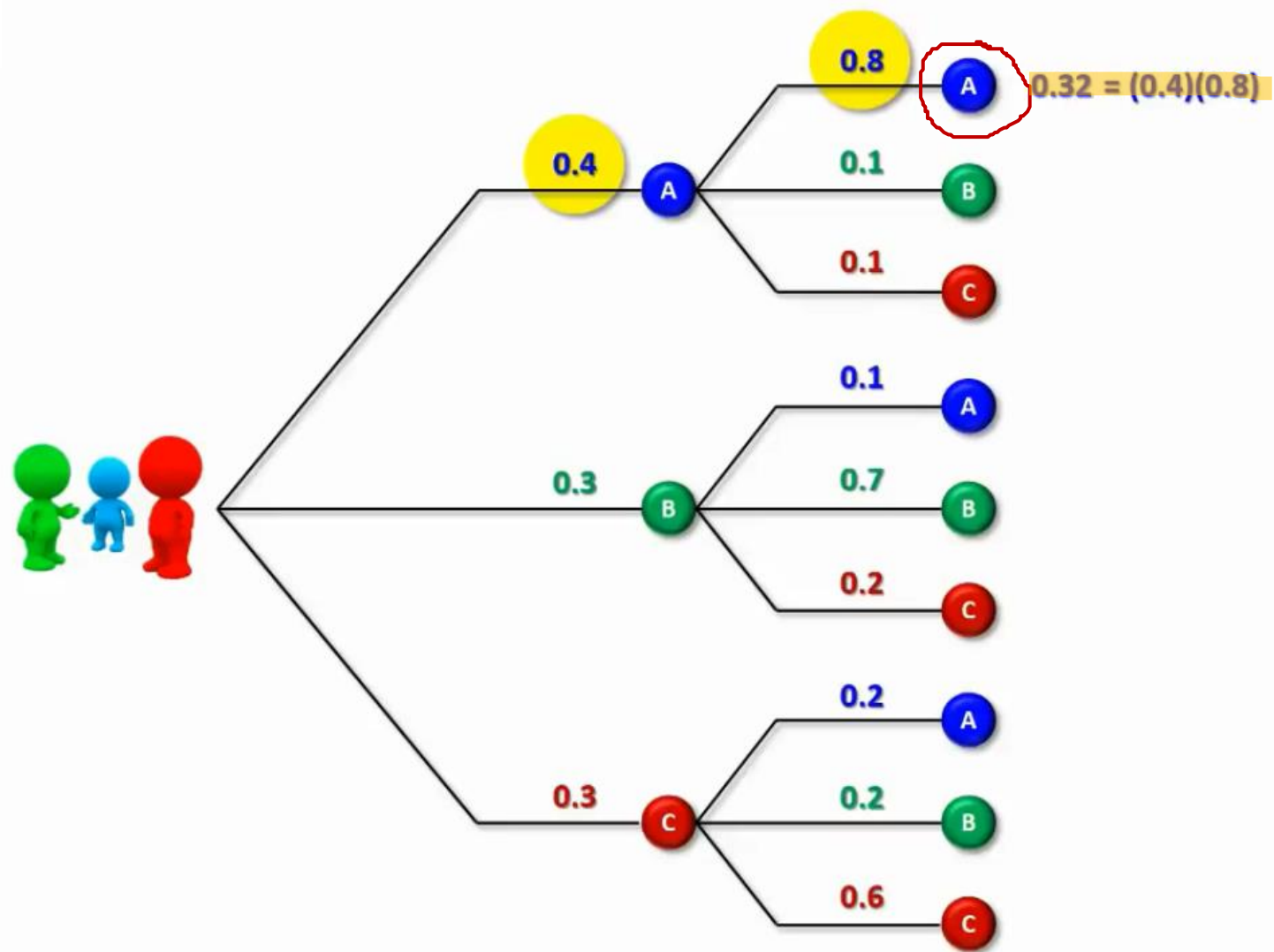
Probabilidades de Estado

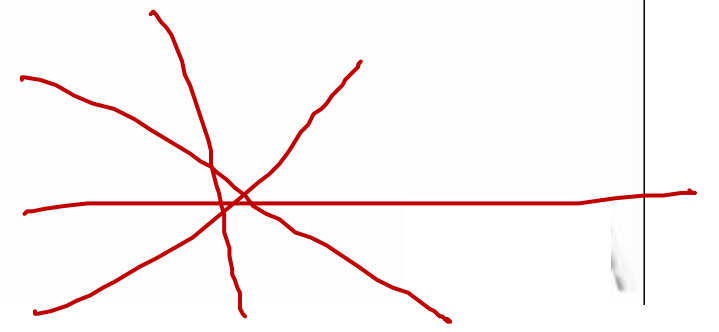
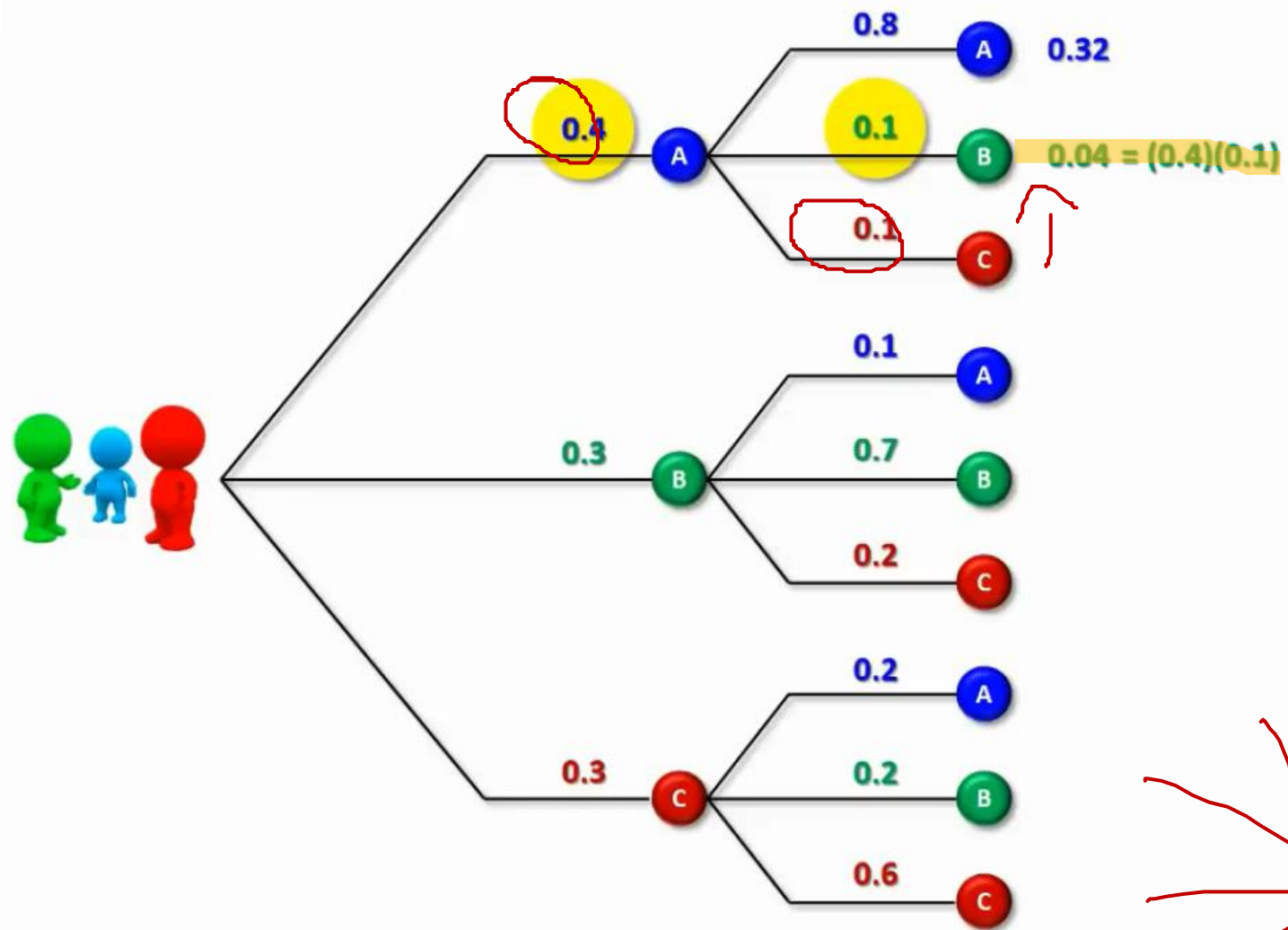


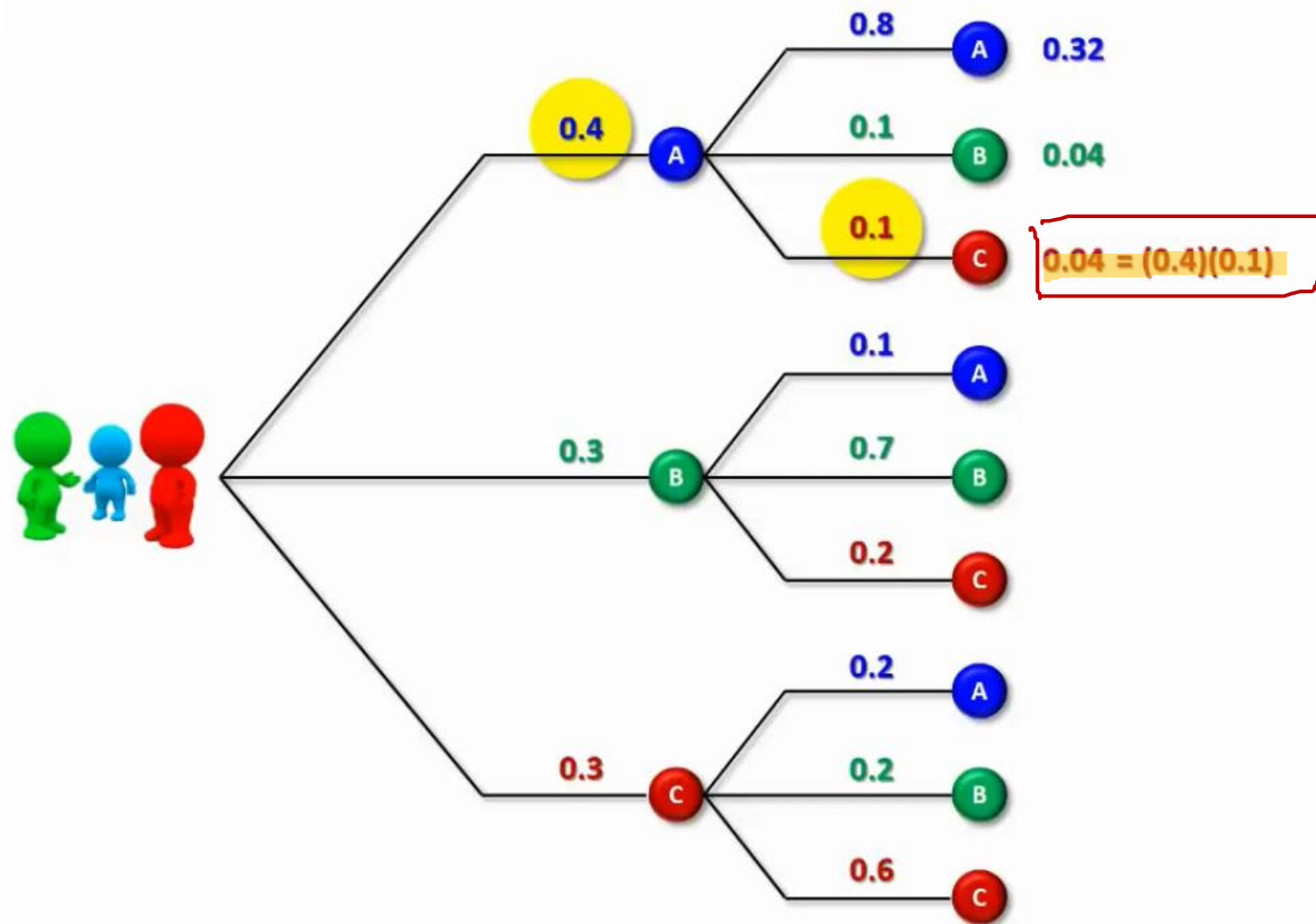
Estado 0

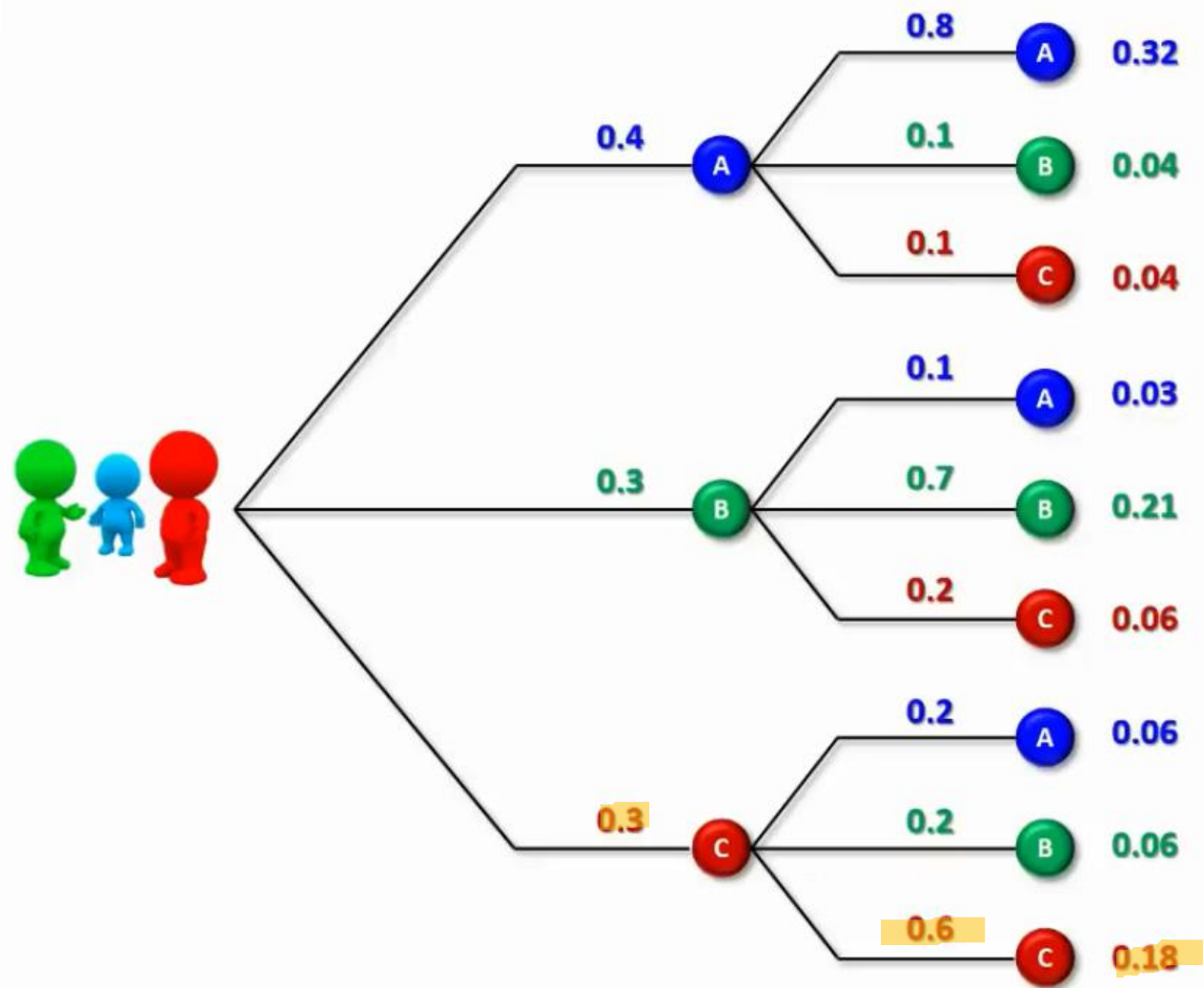
$$\pi(0) = [0.4 \quad 0.3 \quad 0.3]$$

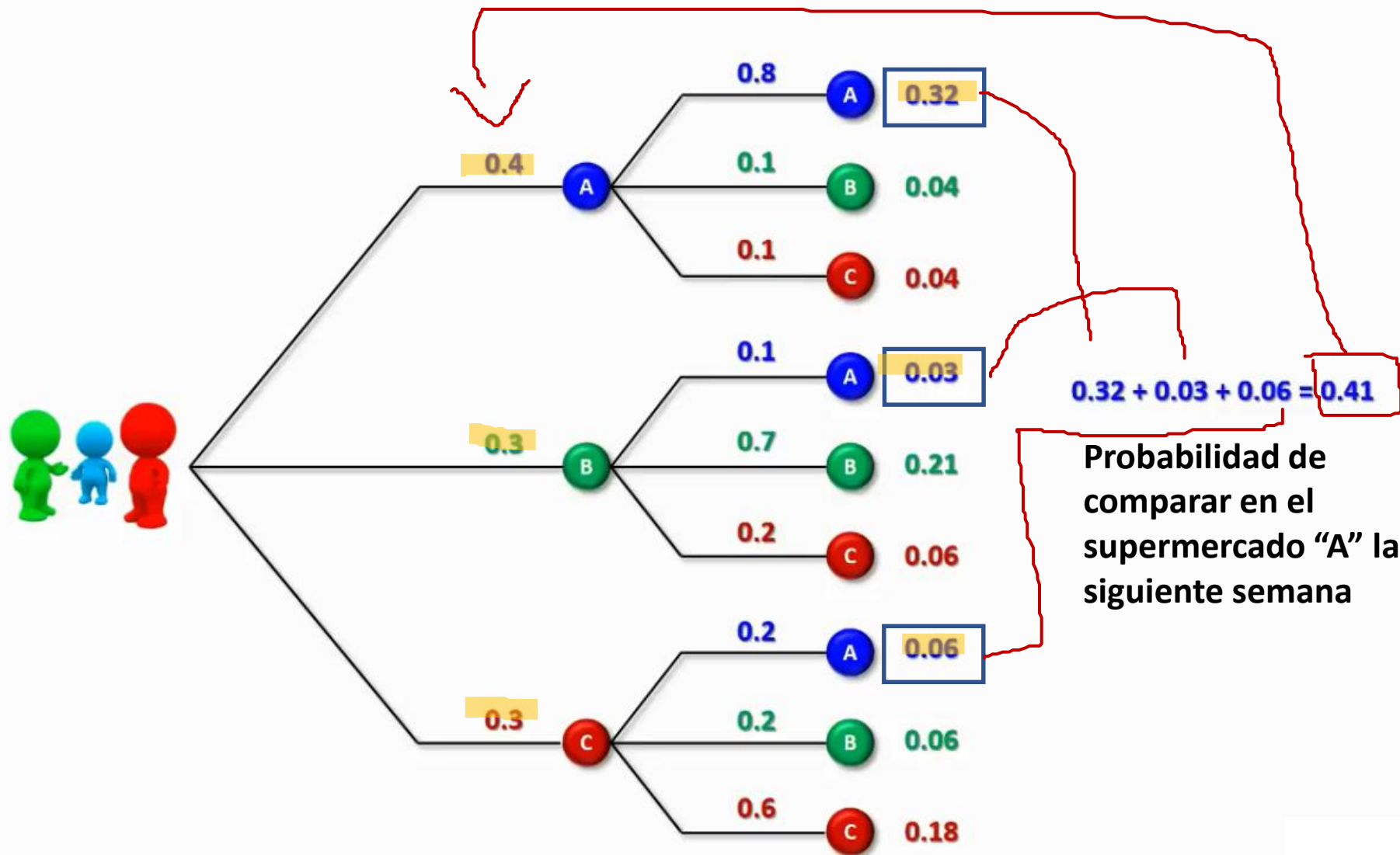
$$\sum \pi_i(0) = 1$$

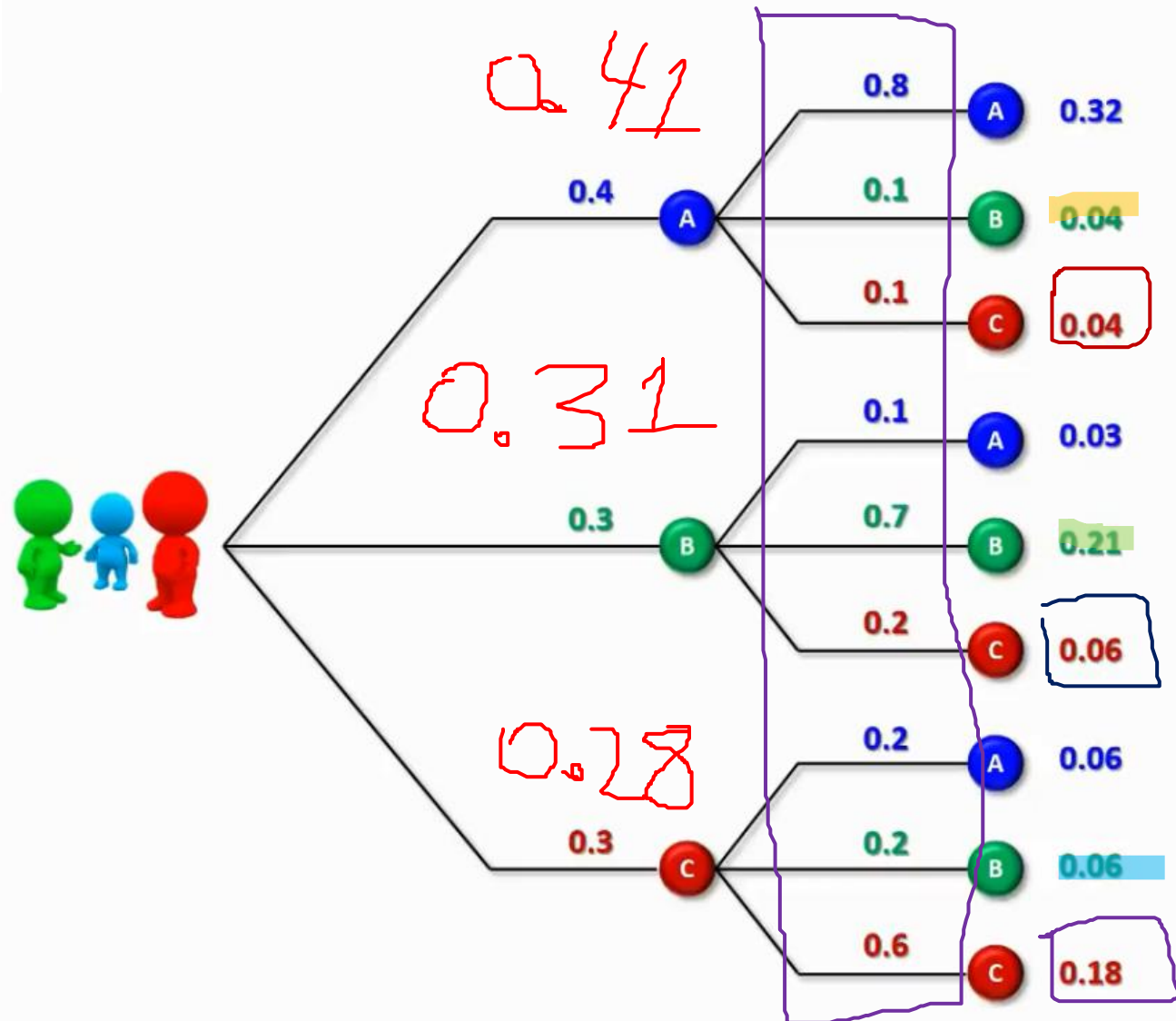








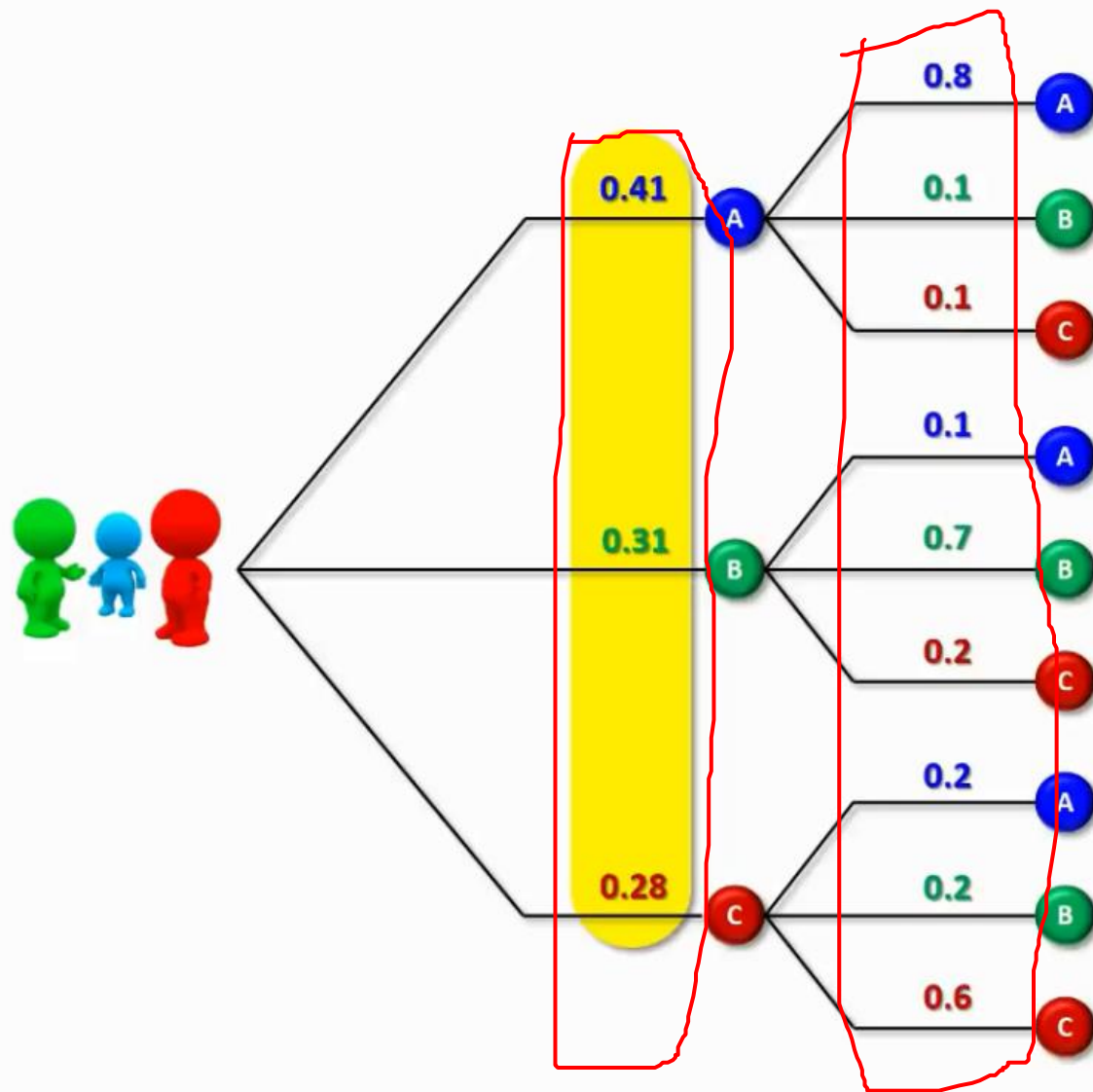




$0.32 + 0.03 + 0.06 = 0.41$ A

$0.04 + 0.21 + 0.06 = 0.31$ B

$0.04 + 0.06 + 0.18 = 0.28$ C

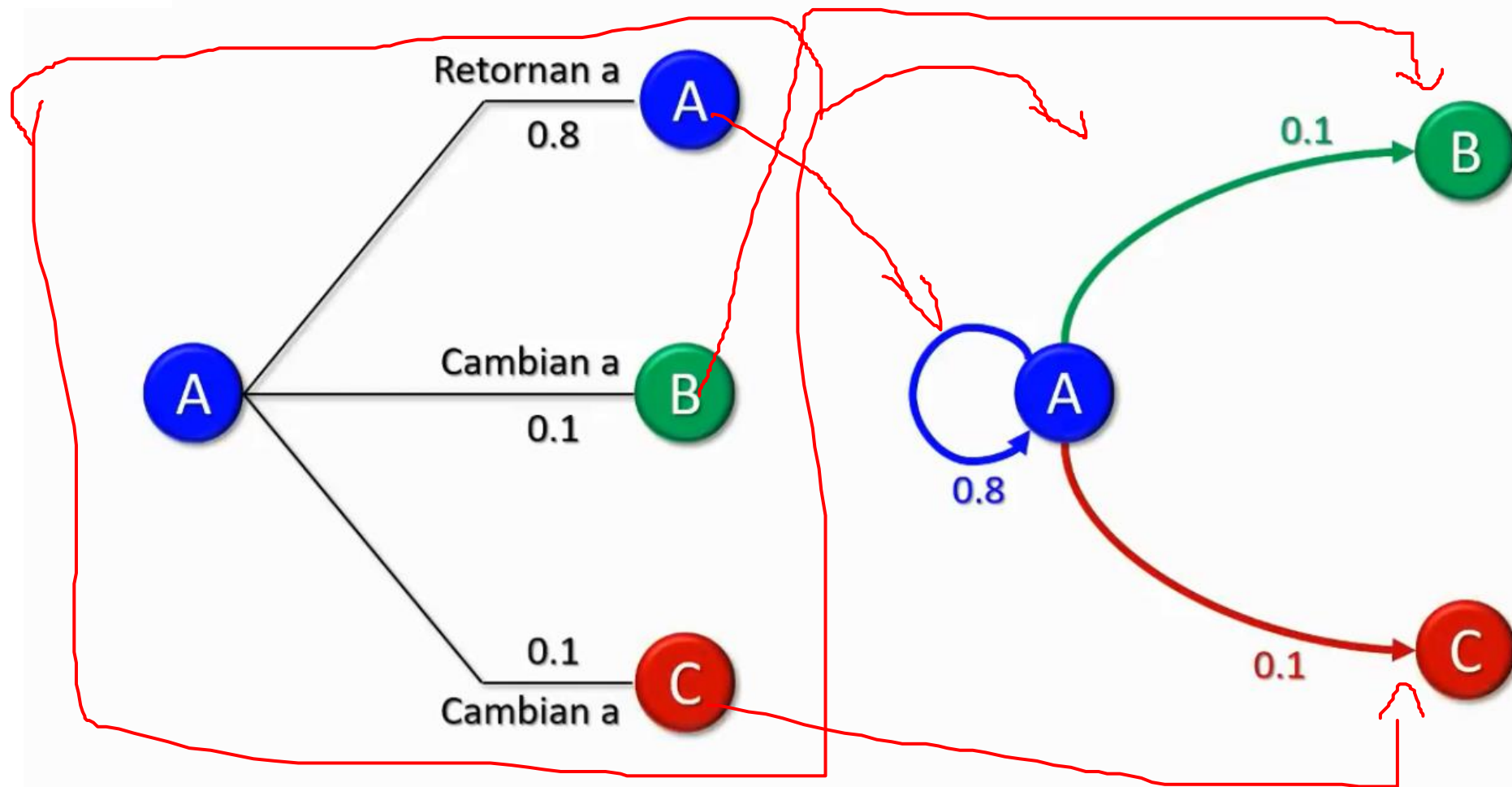


Estado 1

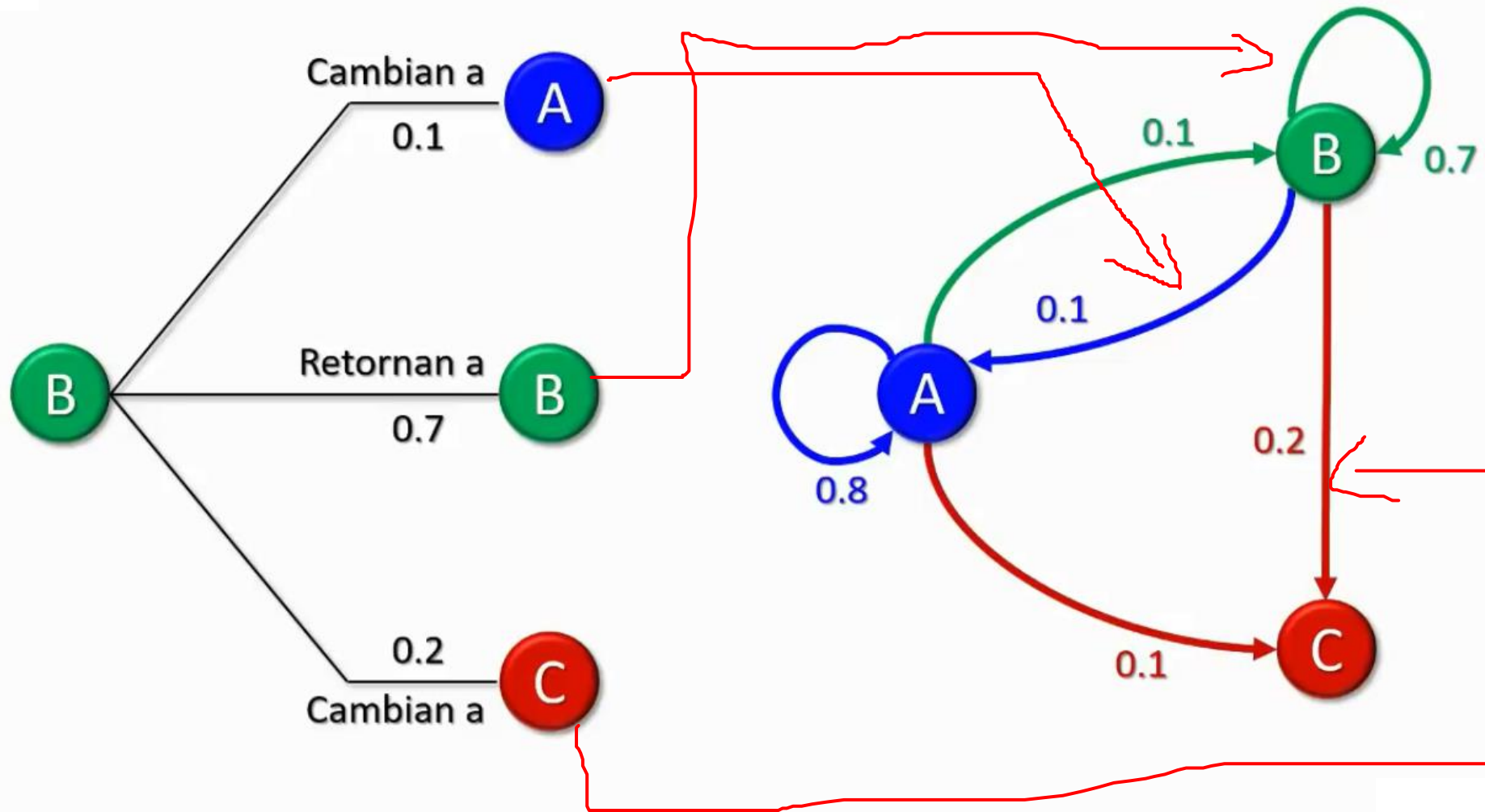
$$\pi(1) = [\underline{0.41} \quad \underline{0.31} \quad \underline{0.28}]$$

Σ 1

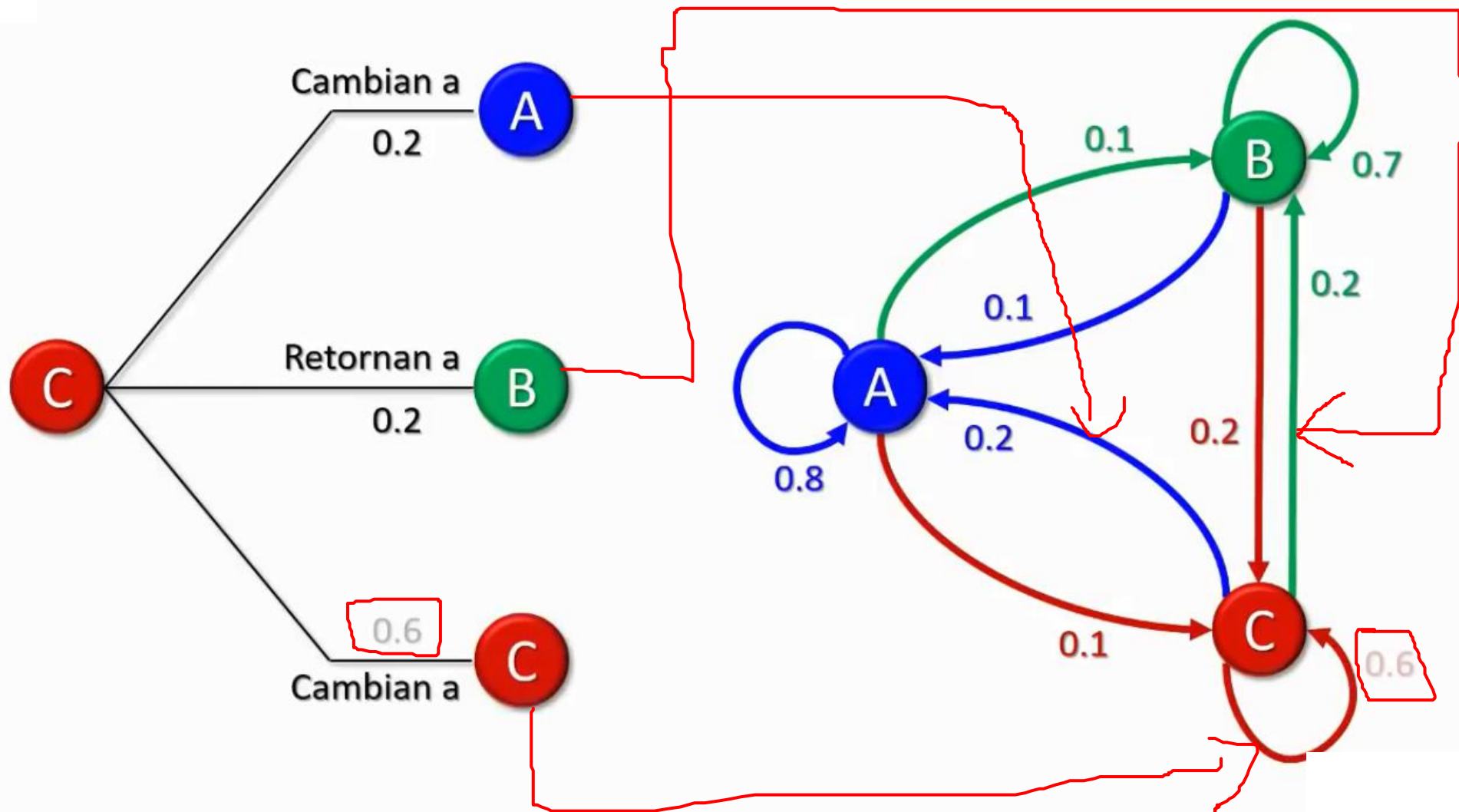
Representación usando Grafos



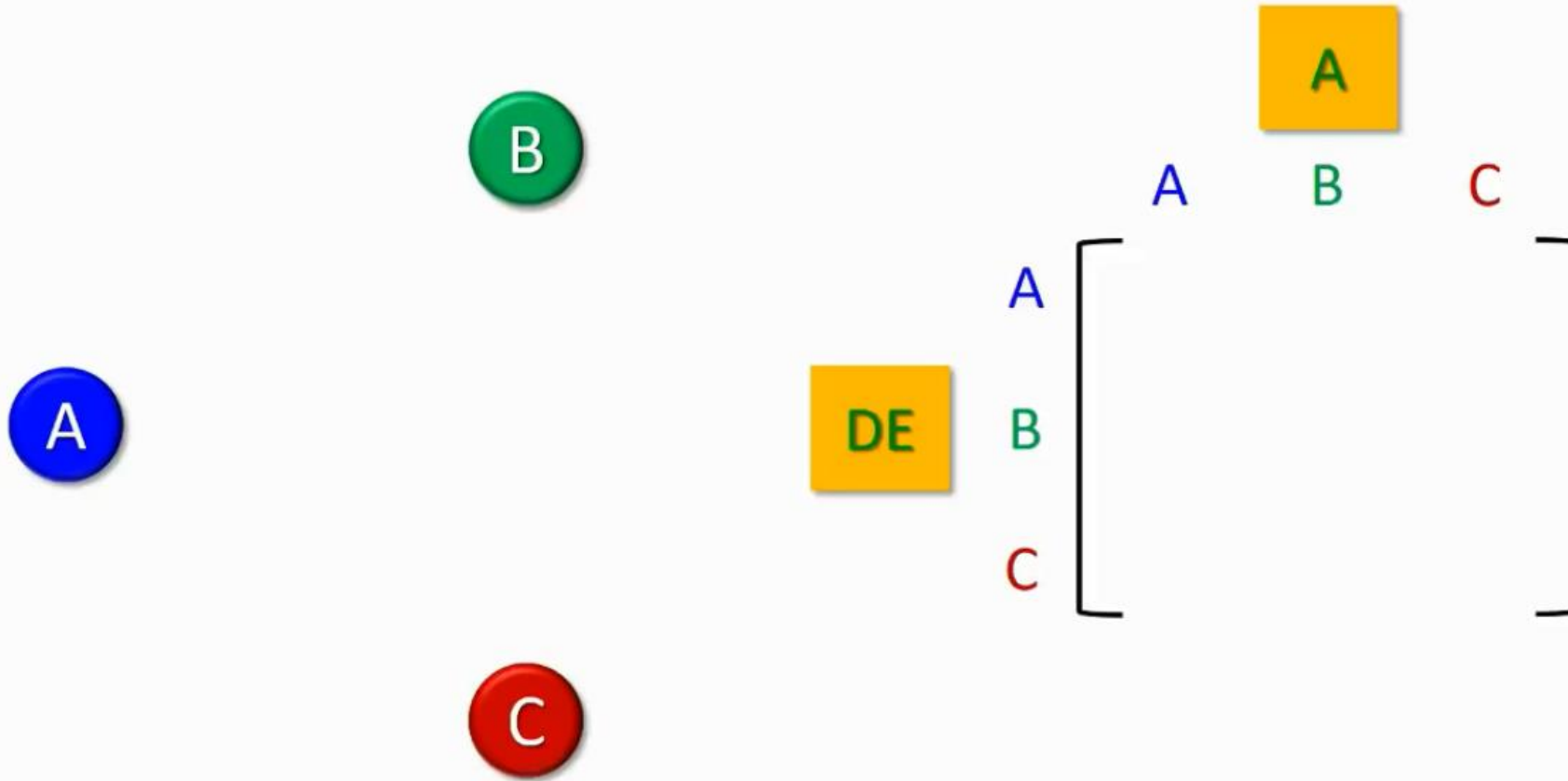
Representación usando Grafos



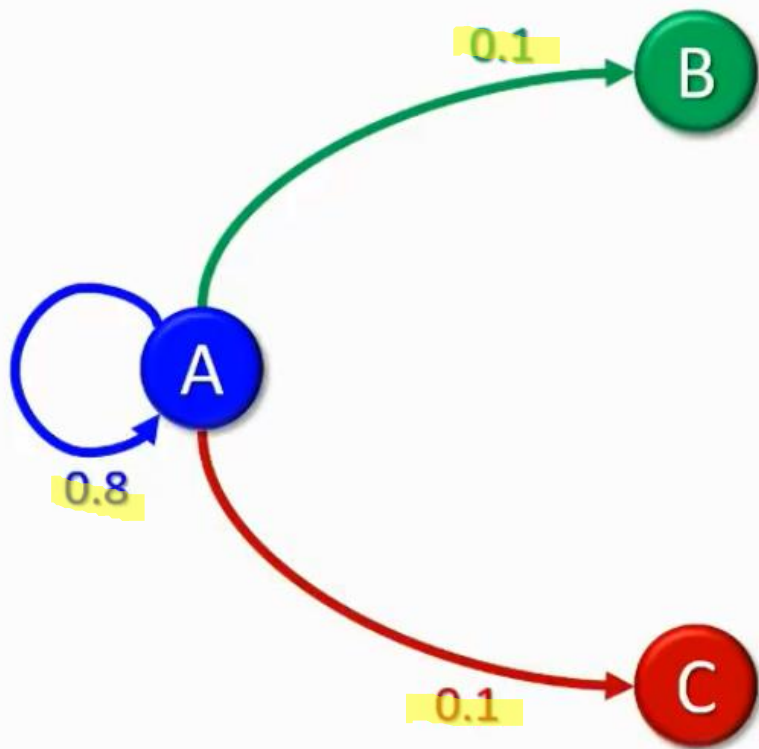
Representación usando Grafos



Grafos a matriz

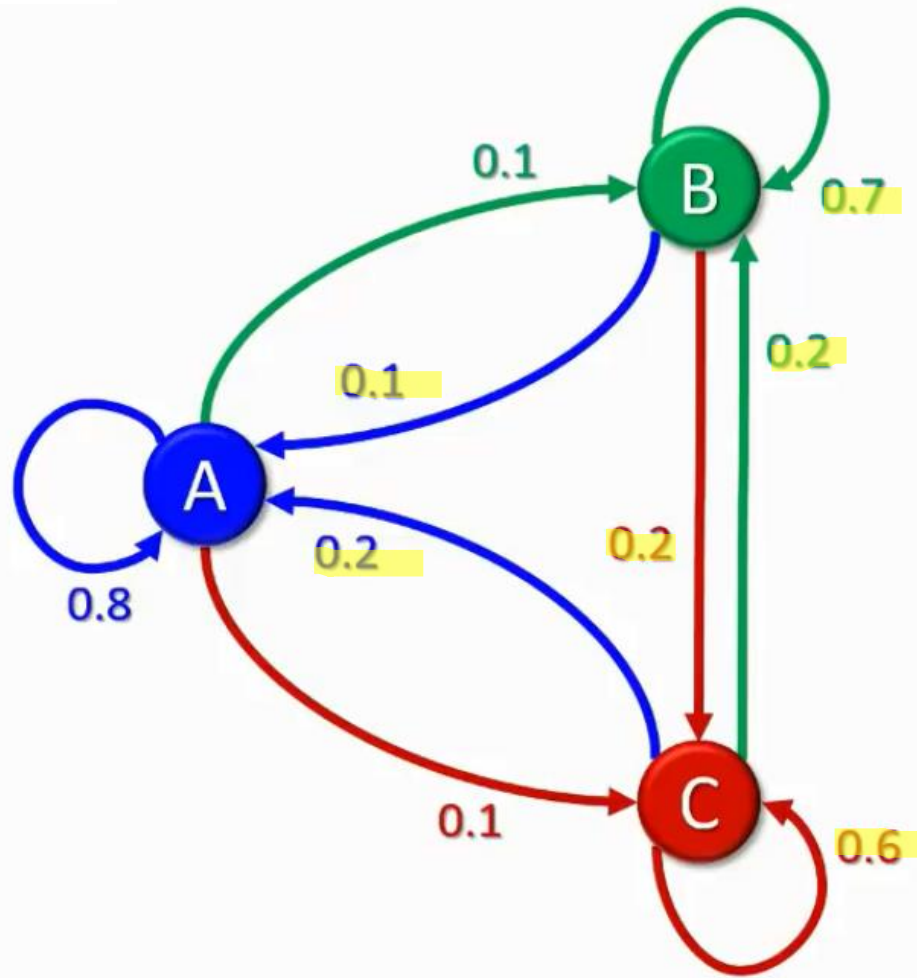


Grafos a matriz



DE	A		
	A	B	C
A	0.8	0.1	0.1
B			
C			

Grafos a matriz



DE

A

	A	B	C
A	0.8	0.1	0.1
B	0.1	0.7	0.2
C	0.2	0.2	0.6

Matriz de probabilidades
de transición

Cálculo de la probabilidad de compra en la siguiente semana usando matriz

$$\pi(1) = \begin{matrix} & \begin{matrix} A & B & C \end{matrix} \\ \begin{matrix} 0.4 & 0.3 & 0.3 \end{matrix} \end{matrix} \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$

$$\pi(1) = \pi(0)P = \begin{bmatrix} \end{bmatrix}$$

Cálculo de la probabilidad de compra en la siguiente semana usando matriz

$$\pi(1) = \begin{bmatrix} 0.4 & 0.3 & 0.3 \end{bmatrix} \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$

$$\pi(1) = \pi(0)P = \begin{bmatrix} 0.41 \end{bmatrix}$$

$$0.4 \cdot 0.8 + 0.3 \cdot 0.1 + 0.3 \cdot 0.2 = 0.41$$

Cálculo de la probabilidad de compra
en la siguiente semana usando matriz

$$\pi(1) = \begin{bmatrix} 0.4 & 0.3 & 0.3 \end{bmatrix} \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$

$$\pi(1) = \pi(0)P = \begin{bmatrix} 0.41 & 0.31 \end{bmatrix}$$

$$0.4 \cdot 0.1 + 0.3 \cdot 0.7 + 0.3 \cdot 0.2$$

$$\pi(1) = \begin{bmatrix} 0.4 & 0.3 & 0.3 \end{bmatrix} \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$

$$\pi(1) = \pi(0)P = \begin{bmatrix} 0.41 & 0.31 & 0.28 \end{bmatrix} \rightarrow \downarrow$$

$$0.4 \cdot 0.1 + 0.3 \cdot 0.2 + 0.3 \cdot 0.6$$

Predicciones a futuro usando Markov

$$\pi(n) = \pi(0) P^n$$

Si deseamos encontrar la tercera predicción nos quedaría:

$$\pi(3) = [0.4 \quad 0.3 \quad 0.3] \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}^3$$

$$\pi(3) = \begin{bmatrix} 0.4 & 0.3 & 0.3 \end{bmatrix} \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}^3$$

Para resolver esto antes debemos encontrar el valor de la matriz al cubo, para ello aplicamos las reglas de potencia de matrices que derivan en multiplicación de matrices. Si tenemos una matriz P^3 para resolverla antes hay que resolver $P^3 = P^2 * P$, lo que nos lleva a encontrar previamente $P^2 = P * P$, por lo cual multiplicaremos la matriz por sí misma y el resultado lo multiplicaremos por la matriz original, hasta ese momento habremos encontrado la matriz P^3

$$P^2 = \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix} * \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$

$$P^2 = \begin{bmatrix} 0.67 \\ \\ \end{bmatrix}^2$$

$$P^2 (1,1) = (0.8 * 0.8) + (0.1 * 0.1) + (0.1 * 0.2) = 0.67$$

$$P^2 = \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix} * \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$

$$P^2 = \begin{bmatrix} 0.67 & 0.17 \\ \vdots & \vdots \end{bmatrix}^2$$

$$P^2(1,1) = (0.8 * 0.8) + (0.1 * 0.1) + (0.1 * 0.2) = 0.67$$

$$P^2(1,2) = (0.8 * 0.1) + (0.1 * 0.7) + (0.1 * 0.2) = 0.17$$

$$P^2 = \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix} * \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$

$$P^2 = \begin{bmatrix} 0.67 & 0.17 & 0.16 \end{bmatrix}^2$$

$$P^2(1,1) = (0.8 * 0.8) + (0.1 * 0.1) + (0.1 * 0.2) = 0.67$$

$$P^2(1,2) = (0.8 * 0.1) + (0.1 * 0.7) + (0.1 * 0.2) = 0.17$$

$$P^2(1,3) = (0.8 * 0.1) + (0.1 * 0.2) + (0.1 * 0.6) = 0.16$$

$$P^2 = \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix} * \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$

$$P^2 = \begin{bmatrix} 0.67 & 0.17 & 0.16 \\ 0.19 & & \end{bmatrix}^2$$

$$P^2(1,1) = (0.8 * 0.8) + (0.1 * 0.1) + (0.1 * 0.2) = 0.67$$

$$P^2(1,2) = (0.8 * 0.1) + (0.1 * 0.7) + (0.1 * 0.2) = 0.17$$

$$P^2(1,3) = (0.8 * 0.1) + (0.1 * 0.2) + (0.1 * 0.6) = 0.16$$

$$P^2(2,1) = (0.1 * 0.8) + (0.7 * 0.1) + (0.2 * 0.2) = 0.19$$

$$P^2 = \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix} * \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$

$$P^2 = \begin{bmatrix} 0.67 & 0.17 & 0.16 \\ 0.19 & 0.54 \end{bmatrix}^2$$

$$P^2(1,1) = (0.8 * 0.8) + (0.1 * 0.1) + (0.1 * 0.2) = 0.67$$

$$P^2(1,2) = (0.8 * 0.1) + (0.1 * 0.7) + (0.1 * 0.2) = 0.17$$

$$P^2(1,3) = (0.8 * 0.1) + (0.1 * 0.2) + (0.1 * 0.6) = 0.16$$

$$P^2(2,1) = (0.1 * 0.8) + (0.7 * 0.1) + (0.2 * 0.2) = 0.19$$

$$P^2(2,2) = (0.1 * 0.1) + (0.7 * 0.7) + (0.2 * 0.2) = 0.54$$

$$P^2 = \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix} * \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$

$$P^2 = \begin{bmatrix} 0.67 & 0.17 & 0.16 \\ 0.19 & 0.54 & 0.27 \end{bmatrix}^2$$

$$P^2(1,1) = (0.8 * 0.8) + (0.1 * 0.1) + (0.1 * 0.2) = 0.67$$

$$P^2(1,2) = (0.8 * 0.1) + (0.1 * 0.7) + (0.1 * 0.2) = 0.17$$

$$P^2(1,3) = (0.8 * 0.1) + (0.1 * 0.2) + (0.1 * 0.6) = 0.16$$

$$P^2(2,1) = (0.1 * 0.8) + (0.7 * 0.1) + (0.2 * 0.2) = 0.19$$

$$P^2(2,2) = (0.1 * 0.1) + (0.7 * 0.7) + (0.2 * 0.2) = 0.54$$

$$P^2(2,3) = (0.1 * 0.1) + (0.7 * 0.2) + (0.2 * 0.6) = 0.27$$

$$P^2 = \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix} * \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$

$$P^2 = \begin{bmatrix} 0.67 & 0.17 & 0.16 \\ 0.19 & 0.54 & 0.27 \\ 0.3 & & \end{bmatrix}^2$$

$$P^2 (1,1) = (0.8 * 0.8) + (0.1 * 0.1) + (0.1 * 0.2) = 0.67$$

$$P^2 (1,2) = (0.8 * 0.1) + (0.1 * 0.7) + (0.1 * 0.2) = 0.17$$

$$P^2 (1,3) = (0.8 * 0.1) + (0.1 * 0.2) + (0.1 * 0.6) = 0.16$$

$$P^2 (2,1) = (0.1 * 0.8) + (0.7 * 0.1) + (0.2 * 0.2) = 0.19$$

$$P^2 (2,2) = (0.1 * 0.1) + (0.7 * 0.7) + (0.2 * 0.2) = 0.54$$

$$P^2 (2,3) = (0.1 * 0.1) + (0.7 * 0.2) + (0.2 * 0.6) = 0.27$$

$$P^2 (3,1) = (0.2 * 0.8) + (0.2 * 0.1) + (0.6 * 0.2) = 0.3$$

$$P^2 = \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix} * \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$

$$P^2 = \begin{bmatrix} 0.67 & 0.17 & 0.16 \\ 0.19 & 0.54 & 0.27 \\ 0.3 & 0.28 & \end{bmatrix}^2$$

$$P^2 (1,1) = (0.8 * 0.8) + (0.1 * 0.1) + (0.1 * 0.2) = 0.67$$

$$P^2 (1,2) = (0.8 * 0.1) + (0.1 * 0.7) + (0.1 * 0.2) = 0.17$$

$$P^2 (1,3) = (0.8 * 0.1) + (0.1 * 0.2) + (0.1 * 0.6) = 0.16$$

$$P^2 (2,1) = (0.1 * 0.8) + (0.7 * 0.1) + (0.2 * 0.2) = 0.19$$

$$P^2 (2,2) = (0.1 * 0.1) + (0.7 * 0.7) + (0.2 * 0.2) = 0.54$$

$$P^2 (2,3) = (0.1 * 0.1) + (0.7 * 0.2) + (0.2 * 0.6) = 0.27$$

$$P^2 (3,1) = (0.2 * 0.8) + (0.2 * 0.1) + (0.6 * 0.2) = 0.3$$

$$P^2 (3,2) = (0.2 * 0.1) + (0.2 * 0.7) + (0.6 * 0.2) = 0.28$$

$$P^2 = \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix} * \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$

$$P^2 = \begin{bmatrix} 0.67 & 0.17 & 0.16 \\ 0.19 & 0.54 & 0.27 \\ 0.3 & 0.28 & 0.42 \end{bmatrix}^2$$

$$P^2 (1,1) = (0.8 * 0.8) + (0.1 * 0.1) + (0.1 * 0.2) = 0.67$$

$$P^2 (1,2) = (0.8 * 0.1) + (0.1 * 0.7) + (0.1 * 0.2) = 0.17$$

$$P^2 (1,3) = (0.8 * 0.1) + (0.1 * 0.2) + (0.1 * 0.6) = 0.16$$

$$P^2 (2,1) = (0.1 * 0.8) + (0.7 * 0.1) + (0.2 * 0.2) = 0.19$$

$$P^2 (2,2) = (0.1 * 0.1) + (0.7 * 0.7) + (0.2 * 0.2) = 0.54$$

$$P^2 (2,3) = (0.1 * 0.1) + (0.7 * 0.2) + (0.2 * 0.6) = 0.27$$

$$P^2 (3,1) = (0.2 * 0.8) + (0.2 * 0.1) + (0.6 * 0.2) = 0.3$$

$$P^2 (3,2) = (0.2 * 0.1) + (0.2 * 0.7) + (0.6 * 0.2) = 0.28$$

$$P^2 (2,3) = (0.2 * 0.1) + (0.2 * 0.2) + (0.6 * 0.6) = 0.42$$

Ahora debemos calcular $P^3 = P^2 * P$

$$P^3 = \begin{bmatrix} 0.67 & 0.17 & 0.16 \\ 0.19 & 0.54 & 0.27 \\ 0.3 & 0.28 & 0.42 \end{bmatrix}^2 * \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$

$$P^3 = \begin{bmatrix} 0.585 \\ \\ \end{bmatrix}^3$$

$$P^3(1,1) = (0.67 * 0.8) + (0.17 * 0.1) + (0.16 * 0.2) = 0.585$$

Ahora debemos calcular $P^3 = P^2 * p$

$$P^3 = \begin{bmatrix} 0.67 & 0.17 & 0.16 \\ 0.19 & 0.54 & 0.27 \\ 0.3 & 0.28 & 0.42 \end{bmatrix}^2 * \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$

$$P^3 = \begin{bmatrix} 0.585 & 0.218 \\ \vdots & \vdots \\ \vdots & \vdots \end{bmatrix}^3$$

$$P^3(1,1) = (0.67 * 0.8) + (0.17 * 0.1) + (0.16 * 0.2) = 0.585$$

$$P^3(1,2) = (0.67 * 0.1) + (0.17 * 0.7) + (0.16 * 0.2) = 0.218$$

Ahora debemos calcular $P^3 = P^2 * P$

$$P^3 = \begin{bmatrix} 0.67 & 0.17 & 0.16 \\ 0.19 & 0.54 & 0.27 \\ 0.3 & 0.28 & 0.42 \end{bmatrix}^2 * \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$

$$P^3 = \begin{bmatrix} 0.585 & 0.218 & 0.197 \\ & & \\ & & \end{bmatrix}^3$$

$$P^3(1,1) = (0.67 * 0.8) + (0.17 * 0.1) + (0.16 * 0.2) = 0.585$$

$$P^3(1,2) = (0.67 * 0.1) + (0.17 * 0.7) + (0.16 * 0.2) = 0.218$$

$$P^3(1,3) = (0.67 * 0.1) + (0.17 * 0.2) + (0.16 * 0.6) = 0.197$$

Ahora debemos calcular $P^3 = P^2 * P$

$$P^3 = \begin{bmatrix} 0.67 & 0.17 & 0.16 \\ 0.19 & 0.54 & 0.27 \\ 0.3 & 0.28 & 0.42 \end{bmatrix}^2 * \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$

$$P^3 = \begin{bmatrix} 0.585 & 0.218 & 0.197 \\ 0.26 & & \end{bmatrix}^3$$

$$P^3(1,1) = (0.67 * 0.8) + (0.17 * 0.1) + (0.16 * 0.2) = 0.585$$

$$P^3(1,2) = (0.67 * 0.1) + (0.17 * 0.7) + (0.16 * 0.2) = 0.218$$

$$P^3(1,3) = (0.67 * 0.1) + (0.17 * 0.2) + (0.16 * 0.6) = 0.197$$

$$P^3(2,1) = (0.19 * 0.8) + (0.54 * 0.1) + (0.27 * 0.2) = 0.26$$

Ahora debemos calcular $P^3 = P^2 * P$

$$P^3 = \begin{bmatrix} 0.67 & 0.17 & 0.16 \\ 0.19 & 0.54 & 0.27 \\ 0.3 & 0.28 & 0.42 \end{bmatrix}^2 * \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$

$$P^3 = \begin{bmatrix} 0.585 & 0.218 & 0.197 \\ 0.26 & 0.451 & 0.197 \\ 0.26 & 0.451 & 0.197 \end{bmatrix}^3$$

$$P^3(1,1) = (0.67 * 0.8) + (0.17 * 0.1) + (0.16 * 0.2) = 0.585$$

$$P^3(1,2) = (0.67 * 0.1) + (0.17 * 0.7) + (0.16 * 0.2) = 0.218$$

$$P^3(1,3) = (0.67 * 0.1) + (0.17 * 0.2) + (0.16 * 0.6) = 0.197$$

$$P^3(2,1) = (0.19 * 0.8) + (0.54 * 0.1) + (0.27 * 0.2) = 0.26$$

$$P^3(2,2) = (0.19 * 0.1) + (0.54 * 0.7) + (0.27 * 0.2) = 0.451$$

Ahora debemos calcular $P^3 = P^2 * P$

$$P^3 = \begin{bmatrix} 0.67 & 0.17 & 0.16 \\ 0.19 & 0.54 & 0.27 \\ 0.3 & 0.28 & 0.42 \end{bmatrix}^2 * \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$

$$P^3 = \begin{bmatrix} 0.585 & 0.218 & 0.197 \\ 0.26 & 0.451 & 0.289 \\ \end{bmatrix}^3$$

$$P^3(1,1) = (0.67 * 0.8) + (0.17 * 0.1) + (0.16 * 0.2) = 0.585$$

$$P^3(1,2) = (0.67 * 0.1) + (0.17 * 0.7) + (0.16 * 0.2) = 0.218$$

$$P^3(1,3) = (0.67 * 0.1) + (0.17 * 0.2) + (0.16 * 0.6) = 0.197$$

$$P^3(2,1) = (0.19 * 0.8) + (0.54 * 0.1) + (0.27 * 0.2) = 0.26$$

$$P^3(2,2) = (0.19 * 0.1) + (0.54 * 0.7) + (0.27 * 0.2) = 0.451$$

$$P^3(2,3) = (0.19 * 0.1) + (0.54 * 0.2) + (0.27 * 0.6) = 0.289$$

Ahora debemos calcular $P^3 = P^2 * P$

$$P^3 = \begin{bmatrix} 0.67 & 0.17 & 0.16 \\ 0.19 & 0.54 & 0.27 \\ 0.3 & 0.28 & 0.42 \end{bmatrix}^2 * \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$

$$P^3 = \begin{bmatrix} 0.585 & 0.218 & 0.197 \\ 0.26 & 0.451 & 0.289 \\ 0.352 & & \end{bmatrix}^3$$

$$P^3(1,1) = (0.67 * 0.8) + (0.17 * 0.1) + (0.16 * 0.2) = 0.585$$

$$P^3(1,2) = (0.67 * 0.1) + (0.17 * 0.7) + (0.16 * 0.2) = 0.218$$

$$P^3(1,3) = (0.67 * 0.1) + (0.17 * 0.2) + (0.16 * 0.6) = 0.197$$

$$P^3(2,1) = (0.19 * 0.8) + (0.54 * 0.1) + (0.27 * 0.2) = 0.26$$

$$P^3(2,2) = (0.19 * 0.1) + (0.54 * 0.7) + (0.27 * 0.2) = 0.451$$

$$P^3(2,3) = (0.19 * 0.1) + (0.54 * 0.2) + (0.27 * 0.6) = 0.289$$

$$P^3(3,1) = (0.3 * 0.8) + (0.28 * 0.1) + (0.42 * 0.2) = 0.352$$

Ahora debemos calcular $P^3 = P^2 * P$

$$P^3 = \begin{bmatrix} 0.67 & 0.17 & 0.16 \\ 0.19 & 0.54 & 0.27 \\ 0.3 & 0.28 & 0.42 \end{bmatrix}^2 * \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$

$$P^3 = \begin{bmatrix} 0.585 & 0.218 & 0.197 \\ 0.26 & 0.451 & 0.289 \\ 0.352 & 0.31 & 0.289 \end{bmatrix}^3$$

$$P^3(1,1) = (0.67 * 0.8) + (0.17 * 0.1) + (0.16 * 0.2) = 0.585$$

$$P^3(1,2) = (0.67 * 0.1) + (0.17 * 0.7) + (0.16 * 0.2) = 0.218$$

$$P^3(1,3) = (0.67 * 0.1) + (0.17 * 0.2) + (0.16 * 0.6) = 0.197$$

$$P^3(2,1) = (0.19 * 0.8) + (0.54 * 0.1) + (0.27 * 0.2) = 0.26$$

$$P^3(2,2) = (0.19 * 0.1) + (0.54 * 0.7) + (0.27 * 0.2) = 0.451$$

$$P^3(2,3) = (0.19 * 0.1) + (0.54 * 0.2) + (0.27 * 0.6) = 0.289$$

$$P^3(3,1) = (0.3 * 0.8) + (0.28 * 0.1) + (0.42 * 0.2) = 0.352$$

$$P^3(3,2) = (0.3 * 0.1) + (0.28 * 0.7) + (0.42 * 0.2) = 0.31$$

Ahora debemos calcular $P^3 = P^2 * P$

$$P^3 = \begin{bmatrix} 0.67 & 0.17 & 0.16 \\ 0.19 & 0.54 & 0.27 \\ 0.3 & 0.28 & 0.42 \end{bmatrix}^2 * \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$

$$P^3 = \begin{bmatrix} 0.585 & 0.218 & 0.197 \\ 0.26 & 0.451 & 0.289 \\ 0.352 & 0.31 & 0.338 \end{bmatrix}^3$$

$$P^3(1,1) = (0.67 * 0.8) + (0.17 * 0.1) + (0.16 * 0.2) = 0.585$$

$$P^3(1,2) = (0.67 * 0.1) + (0.17 * 0.7) + (0.16 * 0.2) = 0.218$$

$$P^3(1,3) = (0.67 * 0.1) + (0.17 * 0.2) + (0.16 * 0.6) = 0.197$$

$$P^3(2,1) = (0.19 * 0.8) + (0.54 * 0.1) + (0.27 * 0.2) = 0.26$$

$$P^3(2,2) = (0.19 * 0.1) + (0.54 * 0.7) + (0.27 * 0.2) = 0.451$$

$$P^3(2,3) = (0.19 * 0.1) + (0.54 * 0.2) + (0.27 * 0.6) = 0.289$$

$$P^3(3,1) = (0.3 * 0.8) + (0.28 * 0.1) + (0.42 * 0.2) = 0.352$$

$$P^3(3,2) = (0.3 * 0.1) + (0.28 * 0.7) + (0.42 * 0.2) = 0.31$$

$$P^3(3,3) = (0.3 * 0.1) + (0.28 * 0.2) + (0.42 * 0.6) = 0.338$$

Volviendo a la fórmula

$$\pi(n) = \pi(0) P^n$$

Tenemos qué:

$$\pi(3) = \pi(0) P^3$$

Habiendo calculado P^3 sustituimos y nos queda:

$$\pi(3) = [0.4 \quad 0.3 \quad 0.3] \begin{bmatrix} 0.585 & 0.218 & 0.197 \\ 0.26 & 0.451 & 0.289 \\ 0.352 & 0.31 & 0.338 \end{bmatrix}^3$$

En las siguientes láminas resolveremos la multiplicación de matrices

$$\pi(3) = [0.4 \quad 0.3 \quad 0.3] \begin{bmatrix} 0.585 & 0.218 & 0.197 \\ 0.26 & 0.451 & 0.289 \\ 0.352 & 0.31 & 0.338 \end{bmatrix}^3 = [0.4176]$$

$$(0.4 * 0.585) + (0.3 * 0.26) + (0.3 * 0.352) = 0.4176$$

$$\pi(3) = [0.4 \quad 0.3 \quad 0.3] \begin{bmatrix} 0.585 & 0.218 & 0.197 \\ 0.26 & 0.451 & 0.289 \\ 0.352 & 0.31 & 0.338 \end{bmatrix}^3 = [0.4176 \quad 0.3155 \quad]$$

$$(0.4 * 0.585) + (0.3 * 0.26) + (0.3 * 0.352) = 0.4176$$

$$(0.4 * 0.218) + (0.3 * 0.451) + (0.3 * 0.31) = 0.3155$$

$$\pi(3) = [0.4 \quad 0.3 \quad 0.3] \begin{bmatrix} 0.585 & 0.218 & 0.197 \\ 0.26 & 0.451 & 0.289 \\ 0.352 & 0.31 & 0.338 \end{bmatrix}^3 = [0.4176 \quad 0.3155 \quad 0.2669]$$

$$(0.4 * 0.585) + (0.3 * 0.26) + (0.3 * 0.352) = 0.4176$$

$$(0.4 * 0.218) + (0.3 * 0.451) + (0.3 * 0.31) = 0.3155$$

$$(0.4 * 0.197) + (0.3 * 0.289) + (0.3 * 0.338) = 0.2669$$

$$\pi(3) = [0.4176 \ 0.3155 \ 0.2669]$$

La probabilidad de A dentro de tres días es 41.76%

La probabilidad de B dentro de tres días es 31.55%

La probabilidad de C dentro de tres días es 26.69%